ISOMETRIC

Vaulted Deep

Project design document

TIRE Project

Vaulted deploys slurry injection technology to geologically sequester organic wastes for the purpose of permanent carbon removal. The TIRE facility is a biosolids injection well co-located with the City of LA's Terminal Island wastewater treatment plant. Over the course of its operations over ten years, the site has safely and effectively injected more than 450,000 tons of biosolids. Vaulted has not yet generated any carbon removal units (CRUs) from TIRE – activities are currently supported through tipping fees from the City of Los Angeles. Vaulted has waived fees for waste volumes over the 5-year baseline to enable the city to send additional volumes to the site. Vault only credits those volumes over the baseline.

Section A: Project setup

Essential project details including who's involved, the location of your project, and how much carbon will be removed.

Protocol requirements

A.1

Projects must provide a brief explanation for why they are eligible under the selected protocol.

SUBMISSION

This Project meets the eligibility criteria for the Biomass Storage in Permeable Reservoirs protocol.

- All feedstocks used in this project accord with the framework set out in the Biomass Feedstock Accounting module v.1.2.
- The project is injecting biomass into a US-based geologic formation for long duration storage via a permitted underground injection well. Further information is provided in the Storage well overview section of this PDD.
- The project is additional, as demonstrated in the section of this PDD covering Additionality.

Ownership

A.2

Projects must provide reasoning and evidence for legal ownership over the rights to all removals that will be claimed.

SUBMISSION

GeoEnvironment currently owns the TIRE injection facility. Advantek Waste Management Services (AWMS) owns the company GeoEnvironment, which holds the rights to all environmental attributes and credits produced from the TIRE injection facility. AWMS and GeoEnvironment hold a contract with the project's waste supplier, the City of Los Angeles, which delineates the City's provision of waste to TIRE. The contract specifies that GeoEnvironment owns environmental attributes and the City cannot generate carbon credits from the waste injected at the TIRE facility. This waste contract with the City will remain in place until 2028, at which point it will be renegotiated. Amendments reflecting new waste

volumes from the City of LA will be registered through a ‘staff note’ that will be made publicly available by the City of LA. Staff notes are a standard way for LA Public Works to modify operational plans.

Vaulted Deep currently owns all rights to carbon credits from TIRE via an offtake agreement with AWMS and GeoEnvironment that confers the right to monetize carbon credits to Vaulted and guarantees Vaulted’s ownership of all carbon credits from the TIRE facility.

Vaulted, AWMS, and GeoEnvironment have negotiated a Transition Agreement that guarantees Vaulted ownership of all environmental attributes for the duration of the Project. Vaulted has also signed an affidavit asserting that it will not enter into any contractual agreement that does not assure Vaulted's ownership of all environmental attributes. If at any point Vaulted does not have full ownership of environmental attributes, the project will be terminated.

FILES

Ownership of Carbon Credits.png, Advantek - Vaulted - Offtake Agreement (Executed) (08.29.2023)(174464355.1) (2).pdf, FINAL FINAL TIRE Contract Amendment 5_Version 11_June2023 signed by OAS (1).pdf, TIRE Transition Agreement Signed 2.5.26.pdf, 2026.02.11AffidavitCreditOwnershipVaulted.pdf, New Credit Ownership Explanation (3).docx, AWMS GET VD 5.7.26 Amendment to Transition Agreement.pdf

Technical description

A.3

Projects must provide a brief technical description of the carbon removal Project activity in accessible language. This should include information on facilities and equipment, the age and average lifespan of equipment, and all further information essential to understanding how carbon removal is achieved by the Project.

SUBMISSION

Vaulted targets moisture-intensive and often pathogenic wastes (biosolids, livestock and agricultural wastes, food waste, paper sludge, etc.) that today are sent to landfills, dumped into waterways, land applied unproductively, or otherwise left to decompose. Utilizing a proprietary slurry injection technology, Vaulted redirects those wastes from being disposed of, minimally processes them, and injects them underground for geologic storage. Vaulted’s slurry injection technology allows for the safe sequestration of solids underground, enabling permanent carbon removal.

Vaulted's TIRE facility is co-located facility with Terminal Island Wastewater Treatment plant. The City of Los Angeles currently pays tipping fees for the biosolids that the city's waste water treatment facilities deliver to the TIRE facility. The project as described in this PDD involves providing additional permanent carbon removal by waiving tipping fees for waste volumes above 1,000 tons of waste per week, thereby attracting additional waste feedstocks from the City of Los Angeles.

At deep well injection sites such as TIRE, Vaulted selects a rock formation with an injection formation at a depth of over 3,000 feet with thickness greater than 50 feet. The depth requirement ensures vertical stress does not propagate horizontally, minimizing risk of fluid migration. The thickness requirement ensures there is sufficiently large waste storage capacity to make a well economically viable. Organic waste is trapped under a confining layer of impermeable rock to ensure containment and long-term storage – these impermeable rock layers are the same types of geologic formations that have kept hydrocarbons underground for millions of years. Vaulted leverages its patented and proprietary well monitoring software to analyze slurry and well characteristics during and after injection. Vaulted uses these insights to optimize its site-specific operational procedures (e.g., injection volumes, batch times, flushing procedures) and ensure subsurface containment on a real-time basis.

The TIRE facility was constructed in 2008 and around November 2008, the first injection activity took place in well SFI#1. The facility has 4 wells, (2-injection wells and 2-monitoring wells), 2 injection pumps (~600HP), pre-injection tank and screener. TIRE currently receives a baseline quantity of biosolids (wetcake) from Hyperion Waste Water Treatment Plant and digested sludge, HPE and brine from Terminal Island. The wetcake is mixed with digested sludge, HPE and brine (depending on availability), slurrified and injected into deep underground formation around 4,900 ft. The permitted injection interval is from 3,800 ft – 7,500 ft, which is a sequence of sand and shale layers.

The project as described in this PDD is not expected to significantly alter business as usual operations at the TIRE facility or require new wells to be constructed; rather it will only increase volumes of waste that the facility accepts. The increased biosolids that the facility accepts will be diverted from land application [redacted]. See 'Baseline' sections below for a description of baseline calculations and crediting thresholds.

FILES

Introduction - TIRE project.pdf

Project participants

A.4

Projects must provide a complete list of organizations participating in the project. This must include: the organization's name, role in the project, registration number, address, contact person, email address, and phone number.

SUBMISSION

Company Registration Number (Unique business identification number in your country of registration):	[redacted]
Organization Name:	Vaulted Deep
Organization Address:	11000 Richmond Avenue, Suite 191, Houston, TX, USA
Contact Person:	Haley Leslie-Bole
Contact Email Address:	haley@vaulteddeep.com
Contact Phone Number:	(925) 639-0453
Organization Role in Project:	Sole owner of environmental attributes and carbon credits

Company Registration Number (Unique business identification number in your country of registration):	[redacted]
Organization Name:	Advantek Waste Management Services
Organization Address:	11000 Richmond Avenue, Suite 190, Houston, TX 77042
Contact Person:	Omar Abou-Sayed
Contact Email Address:	info@advantekwms.com
Contact Phone Number:	(713) 532-7627
Organization Role in Project:	Vaulted shareholder, owner of TIRE facility and owner of subsidiary company GeoEnvironment. AWMS fully owns and operates GeoEnvironment.

Company Registration Number (Unique business identification number in your country of registration):	[redacted]
Organization Name:	GeoEnvironment Technologies

Company Registration Number (Unique business identification number in your country of registration):	**80-0322415**
Organization Address	103 E. Lemon Ave., Ste. 210 Monrovia, CA 91016
Contact Person:	Adri Royal
Contact Email Address:	aroyal@geoenvironment-technologies.com
Contact Phone Number:	(626) 305-0400
Organization Role in Project:	Holder of TIRE environmental attributes

Project locations

A.5

Projects must submit at least one address and/or specific geo-coordinates for the project. Projects may submit multiple project locations – please specify what role each location plays in the project.

SUBMISSION

[redacted]

TIRE facility is in the NW corner of the Terminal Island Water Reclamation Plant, San Pedro, CA with a total acreage of 0.38 acres (1,524.46 sq-meters) as shown in the attached

photograph.

FILES

TIRE Aerial View.png

Removal capacity

A.6

Projects must provide an estimate of the net carbon removal capacity of this project for the duration of the project crediting period (metric tonnes).

SUBMISSION

Year	Estimated carbon removal capacity (metric tonnes)
2026	30607
2027	30607
2028	30607
2029	30607
2030	30607

These quantities reflect facility capacity and waste availability. [redacted]

Section B: Protocol & monitoring data

How will you be measuring, monitoring and maintaining carbon removal, including data models and risk assessments.

Boundaries

B.1

Projects must define the temporal and geographic project boundary.

SUBMISSION

Geographic Boundary:

The TIRE facility is located in Los Angeles, California, USA. In the business-as-usual baseline scenario, waste is currently sourced from two facilities:

1. Digested sludge from digestors at the Terminal Island Water Reclamation Plant through pipeline directly to the mixing pit.

2. Class A biosolids delivered by truck from the Hyperion Water Reclamation Plant located 26.3 miles away at El Segundo, CA.

The TIRE facility currently takes 100% of the sludge from the co-located Terminal Island facility. CRUs will only be generated from additional volumes sourced from the Hyperion facility and other facilities in the future.

Temporal Boundary:

This project will generate CRUs over a period of 5 years and is expected to provide durability of 1,000+ years. While there are no plans to close the TIRE facility in the foreseeable future, Vaulted will undertake monitoring of all sealed wells and has legal obligation to provide financial guarantees for long-term monitoring.

FILES

System Boundary.png

B.2

Projects must define their system boundary and outline all GHGs considered across all sources, sinks and reservoirs (SSRs). Robust justification and appropriate evidence must be provided for any GHG SSRs that have been excluded from the project boundary.

SUBMISSION

Project: All activities undertaken to permanently sequester carbon, at the point of taking ownership of the organic waste are included in the project boundary. Activities within the project boundary include waste transportation, processing, injection, monitoring, and sequestration activities. As a part of these activities, emissions associated with the procurement of all necessary equipment to run the facility for the Project are included, considering standard lifespans. Direct and embodied (upstream) emissions in fuels, materials, and infrastructure are included, as are all emissions associated with site and equipment decommissioning and well plugging. Also included is the counterfactual use case of the organic waste biomass and any relevant replacement emissions, as well as counterfactual carbon storage.

Baseline/counterfactual: All upstream inputs to biomass cultivation and production, as well as all activities tied to the generation of waste, and transportation and other processing activities are included in the project baseline and any increases or changes in emissions resulting from the Project are assessed. Although the Project reduces the number of vehicle miles traveled to transport waste to a disposal facility compared to the Baseline, Vaulted does not credit avoided emissions in alignment with the Isometric Protocol v1.7 Section 2.5.2.

Emissions associated with activities occurring at the TIRE facility that are below the Project crediting baseline are not included in the Project LCA.

Baseline / Project	Carbon Flux / Emission Source	Included/ excluded from project boundary?	Greenhouse Gas(es)	Justification/description
Baseline	Counterfactual Emissions: Fuel Use from Feedstock Sourcing	Included	CO2, CH4, N2O	No additional fuel usage on part of waste supplier due to Vaulted activities. No incremental emissions relative to baseline.
Baseline	Counterfactual Emissions: Electricity Use from Feedstock Sourcing	Included	CO2, CH4, N2O	No additional electricity usage on part of waste supplier due to Vaulted activities. No incremental emissions relative to baseline.

Baseline / Project	Carbon Flux / Emission Source	Included/ excluded from project boundary?	Greenhouse Gas(es)	Justification/description
Baseline	Counterfactual Emissions: Replacement of Feedstock Function	Included	CO2, CH4, N2O	This project meets sustainable feedstock sourcing eligibility criteria and assesses replacement emissions where necessary.
Baseline	Counterfactual Emissions: Temporary Carbon Storage from Feedstock	Included	CO2, CH4	Vaulted calculates counterfactual decay and carbon storage at 15 and 50 years.
Project	Fuel Use from Feedstock Transport	Included	CO2, CH4, N2O	CO2 included as the primary emission from diesel combustion. CH4 and N2O included for completeness.
Project	Embodied Emissions from Feedstock Transport	Included	CO2, CH4, N2O	CO2 included as the primary emission from manufacture of vehicles. CH4 and N2O included for completeness as may also be released during manufacture.
Project	Electricity Use from Biomass Pre-processing	Included	CO2, CH4, N2O	CO2 included as the primary emission from electricity consumption. CH4 and N2O included for completeness as may also be released during electricity generation.
Project	Embodied Emissions from Equipment and Consumables for Biomass Pre-processing	Included	CO2, CH4, N2O	CO2 included as the primary emission from manufacture of equipment and consumables due to electricity consumption. CH4 and N2O included for completeness as may also be released during electricity generation.
Project	Electricity Use from Biomass Injection	Included	CO2, CH4, N2O	CO2 included as the primary emission from electricity consumption. CH4 and N2O included for completeness as may also be released during electricity generation.
Project	Embodied Emissions from Equipment and Consumables for Biomass Injection	Included	CO2, CH4, N2O	CO2 included as the primary emission from manufacture of equipment and consumables due to electricity consumption. CH4 and N2O included for completeness as may also be released during electricity generation.
Project	Carbon Content of Injected Biomass	Included	CO2e	Primary negative emission from process.
Project	Site Decommissioning and Well Plugging	Included	CO2e, CH4, N2O	CO2 included as the primary emission from decommissioning equipment, well plugging and consumables due to electricity consumption. CH4 and N2O included for completeness as may also be released during electricity generation.

B.3

Projects must provide a detailed description of the GHG statement approach and methodology in relation to calculations.

SUBMISSION

Project emissions were calculated based on the activities that took place during the period 12/1/2024 to 12/31/2024. Project emissions and net CO_2e removal will be calculated for every removal activity. Activity data was multiplied by their respective emission factors to obtain the final CO_2e emissions (emissions factors were obtained from the EPA Emissions Factor Library and ecoinvent Database).

Since TIRE has started accepting additional volumes for the purpose of generating CRUs, Isometric's Sampling Method A was conducted, obtaining 30 initial samples. Calculations have since shifted based off Isometric's Method B, where the first 10 trucks of the feedstock will be tested to establish a baseline of carbon content, and every 10th truck is tested to represent the untested trucks.

For each removal activity, emissions were calculated in combination with raw data obtained from the facility and several databases to get the emissions of each respective part of the operation. Energy emissions onsite included the energy required to power operations onsite, which included electricity from the grid and propane. Biosolids Transportation involved CNG consumption, where the quantity was estimated based on distance from the supplier site to TIRE, accounting for inbound and outbound journeys, and the mpg of the delivery truck used for that specific delivery. CNG consumption was then multiplied by the appropriate emissions factor. HPE emissions were calculated based on the assumption the HPE pump works all the time during the daily injection interval. Foreign material emissions were calculated based on the volume of diesel consumption and the associated emissions factor. Pump maintenance also has associated emissions, accounting for any waste residue and cleaner used during maintenance. End of Life (EoL) emissions were calculated based on the equipment used and their corresponding material emissions factor. Lab sampling emissions were calculated based on the weight of the samples and distance to lab.

Total embodied emissions were calculated based on total materials required for the lifetime operation of the project. Given the age of the TIRE facility and its associated equipment, EoL emissions will be the only thing accounted for. Materials were allocated to each metric ton of waste based on a conservative estimate of lifetime processing waste capacity of 1,600,000 metric tons of waste. Emissions associated with each removal activity were calculated based on emissions factors which accounted for the lifecycle impacts of the materials produced.

The city of Los Angeles currently land applies the majority of biosolids from waste water treatment.

[redacted]

Using a two-pool biosolids decay model, which assigns a literature-backed k (decay) rate to a fast (labile) carbon pool and a slow (recalcitrant) carbon pool, Vaulted calculates counterfactual remaining mass of C and CO2e at 15 and 50 years and uses this calculation to identify counterfactual emissions penalties per the Biomass Feedstock Accounting Module v1.2 Section 2.2. This method also takes a conservative approach to estimating the percentage of carbon that is emitted as methane in the labile and recalcitrant pools as well as N2O emissions and uses these to calculate CO2 equivalents.

Net removals in t CO_2e for each removal activity was rounded to the nearest 0.01t CO_2e in line with Isometric requirements.

Emissions factors used were representative of all GHGs.

The GHG statement utilized several resources to obtain the representative emissions associated with the TIRE facility. Energy emissions models from the Los Angeles Department of Water and Power (LADWP) contained electricity share of different energy generation methods. EPA, GREET, and ecoinvent were the emissions factors libraries used for conversion to metric tons of CO2e emitted. Datavault, Vaulted Deep's in-house software, was utilized for LCA calculations. Formulas used in the LCA Calculation can be found in the **TIRE White Paper**. Applications of formula in the Calculations involved with counterfactual decay can be found **here**

Baseline

B.4

Projects must reason and evidence the baseline scenario of their activities having not taken place. Projects will only be credited for removals above this counterfactual baseline.

SUBMISSION

Vaulted sources organic wastes that, in their baseline counterfactual, would have decomposed above ground, re-releasing embodied CO_2e back into the atmosphere. In the case of the TIRE project, the additional volumes of biosolids that the TIRE facility accepts would have predominately been land applied[redacted]

Vaulted and its partner Advantek are currently paid by the City of Los Angeles to receive and store a baseline quantity of biosolids. Vaulted will procure additional biosolids that are currently disposed of via land application by waiving the tipping fee charged to the City of Los Angeles (this is outlined in the Carbon Credit Letter, attached). The increase in biosolids will be used to generate credits, which

represent additional net CDR. However, the baseline quantity of biosolids will not be credited, as this is not additional CDR. Emissions from baseline operations at the TIRE facility are also not included in the LCA because they are not attributable to the carbon crediting Project. To establish a crediting baseline, Vaulted uses historical weekly quantities of biosolids injected for the previous five years, as provided by TIRE, alongside TIRE daily operating logs. The historical baseline will be a simple average of volumes of biosolids injected over the last five years, excluding emergency or Class B volumes. This simple average is appropriate because there is a lack of trend in volumes over the historical 5 year period.

Baseline TIRE activity includes accepting emergency volumes of waste from other LA wastewater treatment facilities. Emergency volumes are typically Class B biosolids, which receive less extensive sanitization treatment than Class A biosolids and are not eligible to be land applied as the City of LA's disposal site. Class B deliveries typically occur when there is an event such as a fire or storm that prevents a wastewater treatment facility from transporting biosolids to the disposal site. Days or weeks where emergency volumes are accepted tend to be significantly higher volumes than non-emergency/ Class B days, introducing outliers to the historical dataset. To manage this, Vaulted excludes emergency volumes from the baseline calculation. Future emergency volumes will also not be eligible to be credited because these events could introduce significant unexpected volumes that are not additional. Where data is missing or there are discrepancies between historical data reporting and daily logs, Vaulted takes a conservative approach by including contested volumes in the baseline calculation to lessen the risk of under-calculating the baseline. The Class A/B status of all loads is tracked in Vaulted's Data Vault system along with any additional emergency designations. Any emergency loads will be noted in Certify and in monthly verification reports if they occur.

Hyperion has provided confirmation that they will accurately label each load of biosolids as Class A or Class B (see email screenshot below). For any load labeled Class B, Vaulted will classify this as an emergency load and eliminate it from creditable volumes. In very rare cases, a truckload of Class A biosolids may be delivered to TIRE under an emergency situation if a truck carrying biosolids is unexpectedly unable to reach an alternative disposal site due to a road closure, weather, or other natural disaster. In these cases, TIRE operators are alerted that an emergency truck is incoming, and they note emergency circumstances in the TIRE daily log and on load tickets. These emergency volumes will also be excluded from creditable volumes. If there is any uncertainty regarding a load's emergency status, Vaulted exclude that load from creditable volumes.

To ensure that there is less than a 10% probability of crediting volumes that are non-additional, Vaulted calculates a 'cushion' volume that is added to the baseline. Qualitatively, the cushion is a factor that is

added to baseline levels to help ensure the additionality of future “additional” volumes, and to protect against over crediting. The 'cushion' is initially a function of Vaulted’s estimates of additional volumes from the City of LA, as well as the variability of historical weekly volumes. The 'cushion' is calculated using simulations of annual volumes, which is based on random sampling of historical weekly volumes and new volumes beginning in the crediting period .

Given the cushion is a function of the actual volumes beginning in the crediting period, not strictly historical volumes, the ultimate baseline (historical volumes + cushion) will be evaluated monthly or quarterly for the first year, and additional, creditable volumes will be based on these incremental assessments, ex-post.

See attached document for additional explanation of the baseline and cushion calculation approach.

FILES

10.14.2025 Carbon Credit Letter Final.pdf, Class B Reporting Screenshot.png, FINAL [External] Vaulted TIRE Baseline Approach_v3 .docx

Leakage

B.5

Projects must evaluate leakage by providing a robust assessment of the potential increases in GHG emissions outside the system boundary that occurs as a result of the project activity.

SUBMISSION

Vaulted assesses all leakage risk across each stage of removal activities, including diversion of nutrients from the city-owned [redacted] where 70% of Los Angeles biosolids are currently land applied for disposal. City of LA staff have confirmed via the attached Waste Partner Form that [redacted] will not increase use of synthetic fertilizer when biosolids are diverted (see attached TIRE Quota Data). [redacted]

City of LA staff also mention in the Waste Partner Form that crop rotations may be altered as a result of Vaulted's diversion of biosolids and overall crop activity may decrease; however, the decrease in total crop output is likely to be minimal and there is evidence that there is regional excess fodder production in the region. The attached analysis confirms that, even under the most conservative estimates, market

leakage of reduced crop production would fall below the materiality limit for carbon accounting. The City of LA's willingness to pay Vaulted a tipping fee for baseline volumes of biosolids also indicates that the City of LA is open to beneficial alternative disposal methods for biosolids.

For these reasons, Vaulted has determined that there are no emissions associated with synthetic fertilizer application and that market leakage from displaced crop production at the [redacted] disposal facility is negligible.

FILES

cba2016-gilbert_jones_healthy_soils_from_biosolids_2016-11.pdf, TIRE Quota Data - 2025.xlsx, Fertilizer Replacement Email Screenshot.png, CoLA Waste Partner Form_482026.docx, [External] Vaulted TIRE Biosolids Market Leakage.docx

Financial additionality

B.6

Projects must demonstrate financial additionality by evidencing removals are the main purpose and only source of revenue; OR demonstrating that economic barriers would prevent project implementation in the absence of carbon finance.

SUBMISSION

[redacted]

Given that the entirety or close to the entirety of TIRE's revenue comes from tipping fees, carbon finance is required to replace lost revenue from tipping fees and cover new expenses incurred to support additional waste volumes [redacted]

[redacted]

Vaulted currently does not qualify for the 45Q tax credit and does not receive any other policy incentive for carbon sequestration.

FILES

10.14.2025 Carbon Credit Letter Final.pdf, Advantek - Vaulted - Offtake Agreement (Executed)(08.29.2023)(174464355.1) (2).pdf, EXTERNAL TIRE Finances.xlsx

Common practice additionality

B.7

Projects must demonstrate that activities similar to the activities of the proposed project are not common practice.

SUBMISSION

Vaulted's patented biomass slurry injection technology has a TRL of 7, defined by the US Department of Energy as a "prototype system demonstrated in an operational environment." The TIRE facility is currently the only facility in the country doing deep well injection of biomass. Vaulted's Great Plains site uses a related but separate approach involving injection into salt caverns.

Environmental additionality

B.8

Projects must demonstrate environmental additionality by evidencing the climate impact of the project is net negative after subtracting the counterfactual CO2 removal and all project GHG emissions, including leakage, from project CO2 removals.

SUBMISSION

Vaulted's injection of biomass waste facilitates the removal of carbon captured from the atmosphere via photosynthesis. The project breaks the carbon cycle by preventing biogenic carbon stored in waste

biomass from being released into the atmosphere as methane and carbon dioxide. In the absence of the project, organic wastes would have decomposed and this carbon back to the atmosphere The project is net-negative within the systems boundaries described below and will durably remove carbon from the atmosphere. A removal activity outlining the associated CO2 stored, emission from operations, and the total emissions is shown below:

CO2e Stored	Energy Emissions, onsite	Biosolids Transportation Emissions	HPE emissions	Foreign Material Transportation, offsite (LA responsibility)	Pump maintenance emissions	End of Life emissions	Lab Sampling, calculations will be adjusted once lab sampling records are obtained	Net CO2e removal
30.735	1.771	0.683	0.027	0.039	0.057	0.019	0.001	28.14

Detailed Calculations can be found within the **TIRE White Paper** and **LCA Excel Spreadsheet** linked

Based on the methodology that encompasses the calculations used to estimate emissions, the project is net-negative within the systems' boundaries described in the GHG statement and will durably remove carbon from the atmosphere.

Regulatory additionality

B.9

Projects must demonstrate regulatory additionality by evidencing that the project is not required by existing laws, regulations, policies, or other binding obligations.

SUBMISSION

This project is not required by any existing laws or regulations. The purpose of the project is to geologically sequester carbon-filled organic waste, resulting in verified carbon removal credits, which can be sold to buyers in the voluntary carbon markets. Therefore, the project meets the Regulatory Additionality requirements of the Isometric Standard.

Note: while not in any way required by existing laws or regulations, the project's storage mechanism is fully regulated as a Class V well. In the US, Vaulted injects under existing permits written for the express

purpose of organic waste injection. These permits are issued by, and regulated under, the EPA or its delegated state agency.

Regulatory compliance

B.10

Projects must assert the method(s) for compliance with regulations for all jurisdictions to which the project is beholden.

SUBMISSION

The TIRE facility is regulated by the EPA under a UIC Class V permit, which was renewed in 2022. The Class V UIC permit is an 'umbrella' permit regulating all drilling and injection activities, so no additional permits are required for increased injection volumes. The site has also undergone review required by the California Environmental Quality Act (CEQA), and the review found no impact or very minimal impact for all project activities (see the attatched CEQA review and findings). Because the TIRE injection facility is located within the Terminal Island Wastewater Treatment Plant, the WWTP's air and water quality permits cover the TIRE injection facility.

Vaulted submits quarterly monitoring reports to the EPA in line with Class V UIC permit requirements. Monitored parameters submitted in these quarterly reports are detailed in E.1.

Permit type	Permit Number	Link
Class V UIC	R9UIC-CA5-FY11-3R CA	https://www.epa.gov/uic/R9UIC-CA5-FY18-1R
Coastal Development Permit	14-03	See attached
NPDES	CA0053856 (Order Nº R4-2021-0095)	https://www.waterboards.ca.gov/losangeles/board_decisions/tentative_orders/individual/npdes/terminal_island/
Air quality permits	Not required	Not required

FILES

Final ApprovedTIRE_CEQA_Comments.pdf, TIRE Coastal Development Permit (2014).pdf

Durability

B.11

Projects must select from the durability threshold(s) defined in the protocol or module to be the project durability threshold.

SUBMISSION

Vaulted confirms that the durability of carbon removed through this project is at least 1000 years, in accordance with the Isometric Standard; we established through direct measurement and modeling of our wells to establish minimum durability, while external literature informs the upper bounds.

Extensive site evaluation combined geological, geomechanical, and fracture modeling to confirm strong containment, stable pressures, and reliable sealing. Ongoing monitoring—including pressure analysis, fiber-optic data, and long-term field observations—shows that injected material remains confined, with no significant migration over more than 15 years. These results supported EPA permit renewal, confirming long-term system integrity.

Because CO_2 diffusion through shale is extremely low, sequestration is effectively permanent as long as wells remain intact or are properly sealed after use. Broader studies suggest storage durability ranging from over 10,000 years to potentially millions of years. For a more detailed geologic and durability assessment report, see page 30 of this **permit**

B.12

Projects must demonstrate a durability in excess of the designated project durability threshold.

SUBMISSION

Vaulted confirms that via our own testing and evidence from the literature, carbon will be securely sequestered for 1,000 years or likely more. We have directly measured and simulated the durability of our own wells to know the minimum durability of our sequestration technology. We rely on external literature to understand the upper bounds on possible durability.

A detailed assessment of the injection formation was conducted through an integrated geological, geomechanical, and fracture simulation approach during the site evaluation and permitting phase. Detailed geological characterization established the stratigraphic continuity, lithologic competency, and sealing integrity of the confining units, while reservoir geomechanical analysis quantified in-situ stresses, pressure limits, and fracture containment thresholds. These evaluations were supported by

fracture simulation model, which predicted the growth behavior and dimensions of induced fractures under anticipated injection pressures and rates. Because novel nature of technology, the facility was initially permitted under a Class V Experimental Permit with a five-year validity period to allow for close performance monitoring and data collection.

Over the course of operations, continuous pressure monitoring and frequent pressure transient analyses (PTA) were used to verify the containment of injected slurry within the targeted formation. The consistent stability of reservoir pressure and absence of pressure communication with overlying formations demonstrated that the injectate remained confined within the intended storage zone. These results provided the EPA Region 9 with confidence in the long-term containment and durability of the system, leading to the renewal of the permit for an additional ten years. This renewal signified regulatory confirmation that the injection formation could reliably maintain integrity and sustain long-term injection without compromising confining zones or nearby wells.

Supporting field data further strengthen this conclusion. The TIRE facility's monitoring wells, originally installed to track pressure propagation, have recorded no significant pressure increase over more than 15 years of injection activity, indicating limited lateral migration of the injectate. Additionally, fiber-optic monitoring confirms that the created fractures remain fully contained within the targeted injection interval. The integration of advanced PTA methods, fiber-optic diagnostics, and fracture modeling conclusively demonstrates that the injection formation exhibits high mechanical integrity, pressure resilience, and long-term durability, ensuring safe, permanent, and stable storage of the injected material

The diffusion coefficient of CO2 (and methane) are exceedingly low through shale, enabling functionally permanent sequestration provided the wellbore itself has integrity or, after its useful life, is properly plugged. Thus, so long as our wells have integrity or, after their useful lives, are property plugged, we can ensure essentially permanent sequestration.

The broader literature gives us an idea of what the duration could be. Simulations of **CO2 geological injection finds durability of 10,000+ years**, while modeled diffusion of CO2 or methane under thick shales suggests ~12 million years of durable storage.

Access to sensor calibration certificates can be found **here**

FILES

Monitoring Requirements.xlsx, Certificate of Calibration for Unit RM-0013.pdf, Certificate of Calibration for Unit RM-0012.pdf, Certificate of Calibration for Unit RM-0011.pdf

Reversals

B.13

Projects must complete the protocol or module specific risk assessment to support the risk of reversal and buffer pool size.

SUBMISSION

#	Question	If answered "Yes"	If answered "No"	Response
1	Is a reversal directly observable with a physical or chemical measurement as opposed to a modeled result?	Proceed to questions 2-9	Proceed to questions 8-9	Yes
2	Is the carbon being stored in an impermeable geologic system? (e.g., salt cavern)	Proceed to questions 8-9	Add 1 to Risk Score and proceed to questions 3-9	No
3	Is the carbon being stored organic?	Add 1 to Risk Score		Yes
4	Are conditions for methane production present (anaerobic conditions, lignin content)?	Add 1 to Risk Score		Yes

#	Question	If answered “Yes”	If answered “No”	Response
5	Does this approach have a material risk of reversal due to natural disasters including, but not limited to, floods, storms, earthquakes, fires, etc.?	Add 1 to Risk Score		No
6	Does this approach have a material risk of reversal due to human-induced events from outside actors, such as change in farming practices, change in ownership and management of Project sites, or similar?	Add up to 2 to Risk Score		No
7	[Applicable only for subsurface storage] Is the carbon being stored with trapping mechanisms preventing reversals? (e.g., multiple confining layers, CO_2 dissolves or solidifies)	Minus 1 to risk (unless 0)		Yes
8	Is there 10+ years of monitoring and/or lab data demonstrating low project risk?	Minus 2 to risk (unless 0)		Yes
9	Does this pathway have a documented history of reversals?	Please consider the frequency and severity of previous reversals, and the shared characteristics between documented reversals and the present project. For pathways with no documented history of reversals, add 0 to the Risk Score. For pathways with a history of frequent reversal, add 2 to Risk Score. For pathways with a limited history of reversals, add 1 to Risk Score.		No
10	Is there one or more Project-specific factors that merit a high risk level?	Please consider the number and severity of risks identified. If one low or medium severity risk is identified, add 1 to Risk Score. If multiple risks are identified, or if any high severity risks are identified, add 2 to Risk Score		No
Total Ris				0

#	Question	If answered “Yes”	If answered “No”	Response
	k Score			

A copy of the Gas Monitoring Data for the verification period can be found here

A copy of the Gas Monitoring SOP can be found here

Uncertainty

B.14

Projects must conduct a sensitivity analysis that demonstrates the impact of each input parameter’s uncertainty on the final net CO_2e uncertainty.

SUBMISSION

Parameter	Data used	Initial assumption	Source / justification	Sensitivity Analysis test: original total net removal (tCO2e), based on 12/01/2024 data	Sensitivity Analysis test: adjusted assumption	Sensitivity Analysis: justification	Sensitivity Analysis test: adjusted net removal (tCO2e), based on 12/01/2024 data	% Change
Transportation	Measured - Fuel use mpg provided by	0.37 gallon/to	Measured data - low uncertainty. An increase of 20% of	28.14	20% increas	Arbitrary increa	28.05	-0.3

fuel use	transport company	n waste	transportation fuel use would not affect results up to the third significant figure (less than 1% in overall total net CO2e removals). Therefore, the original value has been used.		e in fuel use	se tested.		2 %
Distance between biosolids provider and TIRE	Google maps: road miles between supplier and well site	26.3 miles	Measured data - low uncertainty. An increase of 20% of transportation distance use would not affect results up to the third significant figure (less than 1% in overall total net CO2e removals). Therefore, the original value has been used.	28.14	20% increase in transportation distance for biosolids	Arbitrary increase tested.	28.05	-0.32 %
Truck load	Measured at the scale of the biosolids provider	As measured per supplier	Measured data – medium/high certainty. Scales tend to be calibrated annually, with measurements within 100 lb (not confirmed in this case, but many commercial scales work within those limits) . The uncertainty of the measurement would be 0.23% (considering a 20 ton load). Because the uncertainty is well-below the 1% significance threshold, it has not been accounted for.		N/A	Typical scale callibration limits	N/A	N/A
Operation electri	Measured with the electricity reader	As measure	Measured data. Electricity readers typically have an	28.14	2.5 % incr	Electricity Meter	28.10	-0.

city consumption		d per delivery	uncertainty range of +-2.5% (Electricity Meters – Disputed Meter Accuracy Report, National Measurement and Regulation Office, UK). If electricity use was increased by 2.5%, total net CO2e removals would decrease by 0.14%. Because the uncertainty is well-below the 1% significance threshold, it has not been accounted for.		eas e in elec tricity use	s – Disputed Meter Accuracy Report, National Measurement and Regulation Office, UK		14 %
Forklift propane consumption	Calculated from purchase records	3.66 gallons/day		28.14	20% increase in propane consumption	Arbitrary increase tested.	28.13	-0.04%
Site material: steel	Estimated by the LCA expert based on information presented in the site summary document	7.3E-05 t/t waste	High uncertainty, based on expert judgment. An increase of 20% of steel use would not affect total net CO2e removals up to the fifth significant figure (<1%). Therefore, the original value has been used.	28.14	20% increase in steel used	Arbitrary increase tested.	28.14	0.000%
HPE electricity consumption	Estimated, using sludge pump power consumption as a proxy	101 kWh/day	High uncertainty, based on proxies. An increase of 20% of steel use would not affect total net CO2e removals up to the fifth	28.14	20% in electricity used	Arbitrary increase tested.	28.13	-0.04%

			significant figure (<1%). Therefore, the original value has been used.					
Foreign material production	Measured at the site	6000 lbs/week	Medium/low uncertainty, there could be some variations depending on the quality of the feedstock. An increase of 20% of foreign material production would not affect total net CO2e removals up to the thidth significant figure (<1%). Therefore, the original value has been used.	28.14	20% increase in foreign material production	Arbitrary increase tested.	28.13	-0.04%
Pump maintenance emissions	Waste water measured and lubricant calculated from purchase records	30 gallon of waste per week and 0.9 gallon of lubricant per day	Low/medium uncertainty. An increase of 20% of lubricant use and waste water production would not affect total net CO2e removals up to the thidth significant figure (<1%). Therefore, the original value has been used.	28.14	20% increase in lubricant use and waste water production	Arbitrary increase tested.	28.13	-0.04%
Biomass carbon content	TOC % from loads determined through laboratory analysis	6%	TOC % for biosolids data outside of TIRE range from 5-7%. TOC for TIRE is assumed the average of this range and used for this process until real data is obtained					Using Method B to incorporate the average, mean standard error to establish baseline with

								first 10 trucks and every 10th truck after.
GHG emissions of transporting 1 mt over 1 km using a diesel truck, embodied included	Estimated from LCI database	0.000245 73 t CO2e/t*km	High uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.13	-0.04%
GHG emissions of electricity in LADWP	From eGRID data	0.000268 t CO2e/kWh	Low uncertainty. Given the evolution of the CAMX grid emission factor, which has decreased over 15% in the past 4 years it is unlikely that this EF will increase. Anyway, a 20% increase would just decrease net emissions by 1.2%	28.14	20% increase in EF	Arbitrary increase tested.	27.79	-1.24%
GHG emissions of 1 gallon of propane, WTW	GREET 2023	0.00703 t CO2e/gallon propane	Low uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.13	-0.04%
GHG emissions of 1 MJ	GREET 2023	0.000737 t	Low uncertainty. An increase of 20% in the EF would decrease total net	28.14	20% increas	Arbitrary increase	28.05	-0.3

of CNG, WTW		CO2 /MJ	CO2e removals less than 1%		e in EF	tested .		2 %
GHG emissions of transporting 1 mt over 1 km, vehicle and road infrastructure embodied only	customized from ecoinvent 3.10	0.000003807 t CO2e/t*km	High uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.09	-0.18 %
GHG emissions of 1 kg of lubricating oil	proxy from ecoinvent 3.10	0.001766 t CO2 /kg	High uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.14	0.000 %
GHG enissions of 1 kg of waste mineral oil	proxy from ecoinvent 3.10	0.001801 3 t CO2 /kg	High uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.13	-0.04 %
GHG emissions of using the screener during 1 day	customized from ecoinvent 3.10	0.014484895583591 t CO2e/day	High uncertainty. An increase of 20% in the EF would decrease total net CO2e removals less than 1%	28.14	20% increase in EF	Arbitrary increase tested.	28.14	0.000 %
GHG emissions of recycling 1	U.S. Environmental Protection Agency, Office of Resource	0.352739518400326 t	High uncertainty. An increase of 20% in the EF would decrease total net	28.14	20% increase in EF	Arbitrary increase	28.14	0.000 %

ton of steel	Conservation and Recovery (December 2023) Documentation for Greenhouse Gas Emission and Energy Factors used in the Waste Reduction Model (WARM). Factors from tables provided in the Management Practices Chapters and Background Chapters.	CO2e/t	CO2e removals less than 1%	tested.

The change in net CO2e removal induced by each parameter was calculated by varying each parameter independently and observing the impact on net CO2e removals. All the parameters are included in the calculation of the net CO2e removals even if they contribute less than 1% change in the removal. The 1% threshold was applied consistently across all parameters.

B.15

Projects must specify whether they used conservative estimates of input parameters, variance propagation and/or Monte Carlo simulations in consideration of uncertainty (one or multiple options).

SUBMISSION

No	Conservative estimate of input parameters
Yes	Variance propagation
No	Monte Carlo Simulations

Proxies & models

B.16

Projects must describe and justify any models used for quantification, monitoring, and meeting specified protocol requirements.

SUBMISSION

The fracture simulation was conducted using the commercial 3D fracture modeling software [redacted] This study simulated the proposed injection operations in all zones, allowing analysis of fracture dimensions, batch volume, net pressure, and the formation's injection capacity as a function of injection operation and respective zones' geomechanical properties.

Fracture simulation modeling

A 3D fracture simulator is used to evaluate the geometry, growth, and containment of hydraulic fractures within the designated injection zones. The simulator integrates formation mechanical properties such as Young's modulus and Poisson's ratio, in-situ stress contrasts, and fluid/rock interactions to model fracture propagation under varying injection pressures and rates. It provides solutions for estimating fracture height containment, lateral extension, and width distribution while accounting for stress barriers and stratigraphic layering. The model also predicts bottomhole pressure response and fluid efficiency, helping verify that induced fractures remain confined to the target intervals. These outputs are essential for confirming that injected fluids will not migrate into overlying aquifers or offset wells and for demonstrating that the confining layers provide sufficient integrity. By simulating multiple operational scenarios, the fracture model validates containment, optimizes injection parameters, and ensures the protection of underground sources of drinking water.

[redacted], a commercial 3D hydraulic fracture simulation software, was applied to further assess fracture propagation and containment. The program requires key rock and fluid parameters, including Young's modulus, Poisson's ratio, leak-off coefficient, in-situ stresses, injection rates, and fluid rheology (Newtonian or power-law), to characterize fracture behavior. Using linear elastic fracture mechanics coupled with lubrication theory for fluid flow, [redacted] computes how fractures initiate and propagate in layered formations. [redacted] outputs include fracture length, height, width, net pressure, fracture volume, leak-off volume etc., providing a detailed assessment of whether fractures remain within the intended containment zones or pose a risk of extending into adjacent formations

The model is built on assumptions of elastic rock behavior, planar fracture geometry, and simplified leakoff [redacted]), which make it efficient for practical engineering applications. One of [redacted] main advantages is its ability to capture three-dimensional growth trends and provide clear containment assessments for injection planning, making it valuable in both unconventional stimulation and slurry injection studies. While it simplifies geological heterogeneity and fluid complexity, its

structured inputs and well-defined outputs allow for robust sensitivity, giving confidence in predicting fracture behavior under uncertainty.

Sensitivity analysis

Model inputs included in-situ stresses for the injection and containment zones, Poisson's ratio, Young's modulus, leakoff coefficient, fluid properties, and operational parameters such as injection rate and injection duration. These parameters were derived from integrated petrophysical, geomechanical, and in-situ stress analyses. To reduce uncertainty, Vaulted conducted a comprehensive sensitivity analysis by systematically varying key inputs and running multiple fracture simulations. Figure 5 presents a tornado chart illustrating the relative impact of each parameter on fracture area. The chart highlights that leak-off and Young's modulus have the most significant influence, followed by Poisson's ratio and injection rate, while solids concentration and fluid viscosity play comparatively minor roles. With this information, simulation cases that encompassed the best and worst-case scenarios were run. Based on this information, simulation cases were developed to represent both best-case and worst-case conditions. As shown in Figure 6, the best-case scenario (LLH) corresponds to low Young's modulus, low Poisson's ratio, and a high leak-off coefficient, while the worst-case scenario (HHL) corresponds to high Young's modulus, high Poisson's ratio, and a low leak-off coefficient.

To capture the uncertainty embedded within formation properties, Vaulted constructed a case matrix for each injection zone using P10, P50, and P90 values for Young's modulus, Poisson's ratio, and leak-off coefficient, combined with flow rates from 6, 8, 10, to 12 bpm. This ensemble design yielded an average of 108 simulation runs per zone, ensuring robust coverage of the parameter space. The analysis encompassed best-case, average, and worst-case scenarios, ranging from high leak-off and low stiffness to low leak-off and high stiffness. By comparing outcomes across this full range, the simulations demonstrated that induced fractures remain contained within the designated intervals. This systematic approach provides greater confidence in containment, supports the proposed injection rates established in the AOR, and confirms protection of underground sources of drinking water.

The uncertainty analysis is incorporated by systematically varying the three key geomechanical parameters, Young's modulus (YM), Poisson's ratio (PR), and leak-off coefficient (LC), across their P10, P50, and P90 values. These values represent the lower bound (P10), average (P50), and upper bound (P90) estimates derived from petrophysical and geomechanical modeling. By combining these inputs in multiple permutations, the simulations explicitly capture best-case, worst-case, and average fracture responses.

Best case scenario for fracture containment: Achieved when YM and PR are at their lower bounds (P10), and LC is at its upper bound (P90). This combination yields wider fractures with higher fluid leak-off and limited vertical/lateral propagation, resulting in the most favorable containment conditions.

Worst case scenario for fracture containment: Occurs when YM and PR are at their upper bounds (P90) and LC is at its lower bound (P10). This condition produces narrower fractures with reduced

leak-off and greater potential for height growth, representing the most conservative scenario for assessing containment risk.

YM directly corelates with formation breakdown pressure. Higher YM corresponds to higher formation breakdown pressure. Therefore, the worst-case scenario represents the most conservative condition: the injection zone requires the highest pressure to initiate fracturing. Running the model under this condition effectively tests the integrity of the confining zone. If fractures remain contained in the worst-case (HHL) scenario, they are expected to remain contained under all other parameter combinations.

Each geomechanical scenario is also evaluated at multiple injection rates (injection rate range is estimated separately by running Monte Carlo using petrophysical formation properties). Injection rate is a key growth driver therefore, if any combination highlights confining zone breach, the injection rate is reduced and re-tested until fracture containment is achieved. This establishes the upper and lower containment bounds and identifies the operational parameters (rate in particular) that ensures containment under all credible uncertainties.

Across all zones, even the worst-case conditions at optimized rates show fractures fully contained within the designated intervals, confirming protection of the confining units and USDW.

Data collection and storage

B.17

Projects must describe the data collection and storage approach taken, including how data is transmitted, collected and stored, the length of time for which records are archived, backup procedures and strategies and the person(s) / organization(s) responsible for measurement and data collection.

SUBMISSION

Biosolids truck loads are weighed at an external facility using calibrated equipment, and the recorded weights are provided to us regularly along with supporting documentation. No scale is available on site to perform truck weighing. TOC sampling from trucks follows Method B (Sampling a Production Process) in the Isometric's protocol, with three samples taken at different locations within the truck. After an initial period of conducting triplicate measurements for the first 10 batches Vaulted will then proceeded to conduct a triplicate measurement on every 10th batch.

Incoming biomass carbon content is measured at a laboratory, a third-party based in Kansas, following the required standards. Meter readouts of the facility's power consumption can be taken from the

power room, and the facility manager reports this data daily. Power consumption from the pumps used to displace HPE and digested sludge to the site is not quantified in our meter; however, pump specifications are available. The Vaulted site/facility manager will be located at TIRE, and will oversee everything from internal data management, lab analysis, to operational data. ASWM and GeoEnvironment will have no involvement in data collection and storage.

Carbon removal activity data collection and LCA computation is managed on Data Vault, an internally developed data management software developed by Vaulted to collect, track, and analyze CDR data at the TIRE facility. Data Vault enables load by load tracking of all relevant data inputs and calculations for our LCA and, after batched internal approval, automatically shares this data with Isometric for each monthly verification period. Data Vault has been built with scale in mind – it is intended to be leveraged at future sites (both our own and third party owned) and has adequate structure to accommodate different site dynamics. It is now connected to Isometric's dMRV system via API.

Archives for all relevant documentation will be held in compliance with verification and Isometric's protocol of five (5) years and will all to be maintained and backed up through Datavault.

Data Vault serves as the hub not just for data transmission to Isometric but for all stakeholders internal and external. Data Vault enables multiple role type and users according to their specific needs.

Section C: Environmental & social impacts

How will your project affect people and nature, and how will you manage any risks that arise.

Environmental & social impact

C.1

Projects must outline and detail compliance with applicable environmental national and local laws and regulations.

SUBMISSION

The TIRE facility is in compliance with national and local environmental laws and operates under an EPA Class V UIC permit and water and planning permits held by the Terminal Island Waste Water Treatment Plant. The TIRE facility has undergone CEQA review, which evaluates projects for both human and environmental impact. The City of Los Angeles has confirmed that the facility does not require air quality permits.

Permit type	Permit Number	Link
Class V UIC	R9UIC-CA5-FY11-3R CA	https://www.epa.gov/uic/R9UIC-CA5-FY18-1R
Coastal Development Permit	14-03	See attached
NPDES	CA0053856 (Order Nº R4-2021-0095)	https://www.waterboards.ca.gov/losangeles/board_decisions/tentative_orders/individual/npdes/terminal_island/
Air quality permits	Not required	Not required

FILES

TIRE Coastal Development Permit (2014).pdf, Final ApprovedTIRE_CEQA_Comments.pdf

C.2

Projects must provide an overall assessment for the potential material environmental and social impacts, both within and beyond its boundary.

SUBMISSION

Risk Assessment

The TIRE Facility is a fully operational and permitted facility, and risk associated with this project are negligible. Receiving increased volumes of waste from the City of Los Angeles will not significantly impact operations at the facility, or increase the risk of well failure or any other adverse impact. TIRE's operations have been tested for seismic risk and no risk has been identified. Wells at the TIRE facility have also been assessed for their ability to accommodate increased capacity and have been found to have the capacity to accommodate millions of tons of additional waste (see attached).

Environmental Impact Assessment

See attached CEQA EIA completed for the TIRE facility. The assessment found 'Less Than Significant Impact' or 'No Impact' for all evaluated criteria for the TIRE facility, and this project will not change the impact of any of the parameters that were evaluated in the EIA. As stated in the CEQA review, the TIRE site is located in an industrialized area within the Port of Los Angeles and is surrounded by shipping containers, oil and coal storage, and cargo roads so is very unlikely to impact neighboring communities. Neither the project site nor any adjacent areas are included as part of an adopted Natural Communities Conservation Plan (NCCP) or Habitat Conservation Plan (HCP). The CEQA review also finds that the TIRE facility prevents the need for long-distance trucking of biosolids for field application, thereby reducing pollution from trucking in neighboring communities.

FILES

Final ApprovedTIRE_CEQA_Comments.pdf, TIRE Well Properties.pdf

C.3

Projects must demonstrate that it creates no net environmental harm through an environmental impact assessment. This assessment must include, but is not limited to, resource efficiency and pollution prevention and biodiversity conservation and sustainable management of living natural resources.

SUBMISSION

The TIRE Facility underwent environmental assessment prior to securing Class V injection permits. The site also underwent CEQA review (see attachment above). The assessment found no material environmental issue with the site. All necessary pre-injection studies and analyses were conducted before the facility was built, including geologic feasibility studies, local environment and groundwater assessments, and engagement with local community groups and regulators. After the TIRE Facility was fully permitted and operational, the EPA completed a yearlong, in-depth review of the site, resulting in a comfort letter (see attached) citing safe operations and an openness to permitting more of our types of projects.

Pollution Prevention

Increasing waste volumes at the TIRE facility to generate CRUs is not expected to create any additional air or water pollution, increase noise or vibration, or generate any hazardous material. Regular monitoring is undertaken at the facility to ensure that no negative impact occurs (see Monitoring section). Vaulted also maintains several pathways for stakeholders to submit feedback and a protocol for escalating feedback to grievances if any negative impacts are noted by stakeholders (see Stakeholder Input section).

While the project is not expected to generate negative environmental impacts, it has the potential to reduce concentrations of PFAS, nitrates, and other pollutants [redacted]

Biodiversity, Conservation, and Management of resources

This project will have no known impacts on biodiversity or ecosystems. The TIRE facility does not threaten to exacerbate water stress because the facility does not use potentially potable water and site geology has been thoroughly evaluated to ensure that wells will not contaminate groundwater (See Monitoring section).

FILES

PFAS Contamination Article Screenshot.png, EPA Biosolid Comfort Letter Final (1).pdf, EPA Biosolids Pollutants Report.pdf

C.4

Projects must demonstrate that it creates no net social harm by evaluating the potential negative social risks from a project's implementation.

SUBMISSION

Labor Rights	**Vaulted provides a safe and healthy working environment at the TIRE Facility. Standard safety protocol is in place, including daily safety meetings at the start of each shift and providing proper Personal Protective Equipment (PPE). Workers are provided with safety training updated annually based on periodic evaluations of site conditions. Operators are rewarded through their variable incentive compensation for identification of risks and proposing remediations. Moreover, the company provides an anonymous hotline for employees to raise concerns related to any aspect of their work. Additionally, Vaulted pays living wages and commits to paying living wages in the future.**
Land Acquisition and Involuntary Resettlement	The land is owned by the City of Los Angeles. The site is surrounded by the Port of Los Angeles. No known involuntary resettlement happened as a result of the site's development and increasing waste volumes will not trigger any involuntary resettlement.
Impact on Indigenous Peoples, local communities, and cultural heritage	This project has no known impact on Indigenous Peoples, local communities, and cultural heritage
Human rights	While there are not known environmental or environmental justice risks associated with this project, there are major potential benefits associated with Vaulted's activities. As discussed above, diverting land-applied biosolids can prevent PFAS and other harmful substances from entering water and soil.
Stakeholder engagement- Past	At the time of the project's development, the City's outreach program included over 200 community activists, environmental leaders, NGO's, elected and appointed community leaders, regulatory officials, regulatory agencies with jurisdictions over the City of L.A. Bureau of Sanitations (above and beyond the USEPA), and the regional academic communities (see more on engaged stakeholders in D.1 and D.2). Moreover, the City conducted more than 53 site visit tours and community presentations including for the Neighborhood Councils (5), prior to the issuance of the permits and the groundbreaking.
Stakeholder Engagement- Future	Given the site's pre-existing operational status, Vaulted will focus stakeholder engagement on relationship maintenance and continuity of transparency. Vaulted will map all relevant stakeholders and tailor new outreach to stakeholders directly impacted by the project and ongoing operation of the TIRE site. This includes identified high interest/high influence stakeholders, in addition to stakeholders within a 1-mile radius of the project site, which is Vaulted's standard proximal impact radius. The refreshed stakeholder map will directly inform invitations and structure for upcoming virtual office hours in Q4 2026, ensuring that outreach remains inclusive of relevant partners. These sessions will provide an accessible channel for feedback and reinforce Vaulted's commitment to transparency while recognizing that TIRE is an already well-established site that has received past support from stakeholders.

FILES

Community Engagement History.png, Final ApprovedTIRE_CEQA_Comments.pdf, Duplicate_of_TEMPLATE_Stakeholder_Register_1759525450.xlsx, Exemption Note_ TIRE April 2026 Validation and Verification.docx

Sustainable development

C.5

Projects must demonstrate how their carbon removal activities are consistent with relevant SDGs.

SUBMISSION

#SDG with name	Description of project impact (as per selected indicators corresponding to the targets (example) predefined under each SDG by the UN)	Description of alignment with indicators of jurisdiction(s), if relevant and applicable	If applicable, unit of measurement (e.g. removal credits, MWh)
#6 Clean water and sanitation	This project aligns with goal 6.3: "By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally" and indicator 6.3.2 " Proportion of bodies of water with good ambient water quality"	By diverting biosolids [redacted] Vaulted prevents [redacted] contaminants from entering ground and surface water, or 'ambient water'. This also prevents nitrate leaching into drinking water. [redacted] Vaulted plans to work with Academic and NGO partners to better quantify the non-point source water quality benefits of Vaulted's interventions.	PFAS chemical concentration, (parts per million); nitrate concentrations (parts per million)

Project closure

C.6

Projects must describe the conditions under which the project will be considered closed, and describe the project closure plan – outlining any post-cessation actions that will be undertaken upon closure of the project.

SUBMISSION

The TIRE Project will be considered closed when Vaulted communicates its intention to cease carbon credit generation at TIRE. While Project closure will ideally correspond with the closure of active injection wells, it is possible that well closure and Project closure will not occur simultaneously. If wells remain open after Project closure, Vaulted will take the relevant steps to ensure that it can monitor wells for permanence and/or safely plug and abandon wells at a later date as described below.

If well plugging and abandonment does occur concurrently with Project closure, before initiating any well abandonment activities, Vaulted will notify the EPA at least 60 days in advance and wait for written approval before proceeding. All plugging and abandonment work will be performed in accordance with the EPA-approved Plugging and Abandonment Plan consistent with CALGEM's Onshore Well Regulations and 40 CFR § 146.10. If the well design or operating conditions change, Vaulted will revise the plan as directed by the EPA to ensure continued compliance. To maintain adequate financial assurance, Vaulted will obtain and periodically update a third-party cost estimate covering all materials, labor, and services required to properly plug and abandon each well in full accordance with the approved plan.

If any well remains inactive for more than two years and is not being used as a monitoring well, Vaulted will proceed with plugging and abandonment in accordance with the approved plan, unless an EPA-approved plan is in place to maintain the well safely. During periods of inactivity, Vaulted will perform annual external and internal mechanical integrity tests (MITs) every two years to verify well integrity and ensure protection of underground sources of drinking water (USDWs). Should any loss of mechanical integrity occur, Vaulted will immediately take corrective actions to restore it before reactivation or continued idling of the well.

Within 60 days after completing any plugging operation, or with the next quarterly report (whichever comes first), Vaulted will submit a Plugging and Abandonment Report to the EPA using Form 7520-19, along with the engineer's log, daily rig log, and certification of the work performed. The report will confirm that the plugging was conducted in accordance with the approved plan or provide justification for any procedural deviations. In the event of complete facility cessation, Vaulted will notify the EPA

and submit a comprehensive decommissioning plan that details the plugging of all wells, removal of associated pipelines and surface equipment, and restoration of the site. This plan will ensure that all operations are concluded safely and that the facility poses no future environmental or groundwater risk following closure.

The TIRE facility is located within the premises of a wastewater treatment plant; therefore, no permanent pits, anchored tanks, or dedicated tank farms have been constructed. All surface equipment, including piping, pumps, and frac tanks, are modular and designed for reuse.

At the end of the facility's operational life, a comprehensive decommissioning plan will be implemented. This plan will include the safe removal and recycling of all temporary equipment and frac tanks, proper flushing and cleaning of lines, and restoration of the site to its original condition in accordance with applicable regulatory requirements. The goal is to ensure that all materials are either repurposed for future use or responsibly disposed of, leaving no environmental footprint once the injection operations are concluded.

At TIRE, we run reservoir simulations to estimate gas migration associated with our injection operations. We use [redacted] a numerical simulator that models how fluids, gases, and heat move through underground rock over time by solving coupled mass- and energy-balance equations. Since, TOUGH3 can represent multiphase flow, pressure buildup, and plume migration in relatively complex and layered subsurface systems, it can be used for predictive estimations.

The simulation model is updated and run every quarter, and results to date show that the plume remains relatively close to the injection well. This is consistent as the monitoring wells, have not detected any changes in pressure or temperature, that suggest plume is relatively closer to the injection wells and not reached the monitoring wells. Also, TOUGH3 can be used to predict how the plume will behave after injection operations cease at TIRE, supporting evaluation of long-term subsurface performance and containment.

Section D: Stakeholder input process

How will your project work with and respond to the community while staying compliant and adaptable.

Stakeholder consultation

D.1

Projects must provide a description and documentation of how comments by local stakeholders have been invited and compiled, a summary of comments received, and report on how due account was taken of comments received.

SUBMISSION

1. Purpose and Objectives

Vaulted is committed to inclusively and transparently engaging communities and stakeholders at the TIRE facility to ensure that site operations and biomass sourcing activities are informed by local feedback. The TIRE site has been in operation for close to a decade, and significant community engagement efforts were conducted when the facility received its Class V permit and underwent CEQA review (see attached slide). Vaulted will continue to expand community engagement efforts on the ground in alignment with Isometric Standard Section 3.5 while solidifying our own internal community engagement best practices.

This plan outlines planned stakeholder mapping and engagement to update the facility's stakeholder activities.

2. Stakeholder Identification

Past efforts:

At the time of the project's development, the City's outreach program included over 200 community activists, environmental leaders, NGO's, elected and appointed community leaders, regulatory officials, regulatory agencies with jurisdictions over the City of L.A. Bureau of Sanitations (above and beyond the USEPA), and the regional academic communities. A sample of the groups included in the outreach program included:

- Santa Monica Bay Keeper (now WaterKeeper)
- Heal-the-Bay
- The Sierra Club
- The NRDC
- SM Conservancy

- LA Conservation Corp
- The Port Citizens Advisory Committee (since disbanded and reorganized into other groups)
- Environmental Defense Council
- Climate Resolve

Moreover, the City conducted more than 53 site visit tours and community presentations including for the Neighborhood Councils (5), prior to the issuance of the permits and the groundbreaking.

Both the CEQA and the Class V injection well permits also went to public notice with open comment periods. During the CEQA process, the Bureau of Sanitation, as lead agency under CEQA, committed to a full public distribution and comment process including to public agencies and other interested parties and organizations. The CEQA process also included opportunities for members of the public and agencies to address both the Board of Public Works and the City Council during hearings before the ND was adopted.

Following this outreach and public comment periods, the Board of Public Works reviewed the CEQA Negative Declarations together with all received comments and then made a recommendation to the City Council as the final decision-making body.

Future Engagement:

Vaulted will undertake additional stakeholder mapping and engagement in Q1 of 2026. Vaulted uses a systematic and replicable stakeholder mapping process during early project development. This includes:

- Conducting preliminary research on local environmental, economic, demographic, and political conditions.
- Defining geographic scope based on regional ties and environmental vulnerabilities.
- Identifying stakeholders through desktop research, local media, consultants, landowner meetings, and personal connections.
- Categorizing stakeholders by type (e.g., community members, government bodies, NGOs, educational institutions).
- Special focus is placed on fenceline stakeholders and stakeholders directly impacted by the project.

Given the sit's pre-existing operational status, Vaulted will focus stakeholder engagement on relationship maintenance and continuity of transparency. Vaulted will map all relevant stakeholders and tailor outreach to stakeholders directly impacted by the project and ongoing operation of the TIRE site. This includes identified high interest-high influence stakeholders, in addition to stakeholders within a 1-mile radius of the project site, which is Vaulted's standard proximal impact radius. Given the site's location within a secure, industrial setting, these mapping efforts will include institutional and operational stakeholders as well as stakeholders potentially materially impacted by project activities.

The refreshed stakeholder map will directly inform invitations and structure for upcoming virtual office hours, ensuring that outreach remains inclusive of relevant partners while appropriately scoped to the project's context. These sessions will provide an accessible channel for feedback and reinforce Vaulted's commitment to transparency while recognizing that TIRE is an already well-established site that has received past support from stakeholders.

3. Engagement Methods and Frequency

Public comment was solicited both during the CEQA review process and the Class V UIC permitting process. Comments and responses can be found at the end of the attached documents (will attach Class V and CEQA docs). Neither comment period yielded significant objections to the facility, but stakeholders did attest to their support for the facility.

To ensure that stakeholders are able to learn about and provide input into this new project, Vaulted will host virtual office hours in the spring of 2026. A virtual format will allow a broad range of stakeholders to participate. All relevant stakeholders will be given 14-day notice before office hours or other events and translation services will be available where necessary. Invitations will be broadly distributed to ensure inclusivity. Vaulted will record and submit any feedback received during this session via our online feedback log. For those not able to attend virtual office hours, Vaulted will respond to questions over email and have 1-1 meetings where appropriate.

The following table outlines Vaulted's planned stakeholder engagement activities:

Engagement Activity	Frequency	Stakeholder Group	Objective	Responsible Party
Office hours	Annual, starting in Q4 2026	Institutional, regulatory, and operational stakeholders identified in the updated map	Maintain transparency; provide a forum for feedback and questions	External Affairs Team

Engagement Activity	Frequency	Stakeholder Group	Objective	Responsible Party
Targeted email updates	Semi-annual or as needed	High interest stakeholders identified in the updated map, in addition to others who might opt in to these updates	Share brief project updates and maintain open communication channels	External Affairs Team
In-person meetings	As needed	High interest stakeholders identified in the updated map, in addition to others as needed	Address technical or compliance-related topics and sustain working relationships	External Affairs Team
Feedback & Grievance Mechanism log review	Weekly	Submitters	Track and resolve submissions as necessary	External Affairs Team

4. Input Collection and Tracking

In order to solicit feedback from all potential community stakeholders, all Vaulted sites have a site-specific email address launched during the project announcement and administered by External Affairs team members. Vaulted will create a site-specific email address and phone number for the TIRE facility. Stakeholders are also able to submit feedback via Vaulted's website, where the team will add an option to characterize feedback as a 'grievance'.

Since Vaulted's External Affairs team is small and community engagement needs are growing quickly, we have expanded and formalized our procedures for tracking stakeholder feedback. Vaulted recently stood up its stakeholder input log using the project management software Monday.com. Stakeholder maps will be built in the input log and then used to document ongoing interactions as well as incoming engagement.

The types of feedback recorded in the stakeholder input log will include concerns received via email interactions, outcomes of External Affair's stakeholder meetings, and questions received from our online contact form. As a small team, we often rely on staff integrated within the community on the ground. Every month, External Affairs plans to send an email to local staff and contractors requesting reports of any community concerns or other sentiments they have observed. The email will also remind them of the availability of the grievance mechanism.

The input log documents key details about the stakeholder, our relationship with them, preferred communication methods, as well as their input and sentiment toward the project. The input log is also integrated with our grievance mechanism process in that it tracks our timelines for acknowledgement, resolution, and closure status. This will enable both tracking of feedback and escalation of grievances

to be monitored and reported in a more robust and organized way. See attachment for a mockup of our Monday Stakeholder Input Log in Excel.

5. Response and Follow-Up

As demonstrated in the Stakeholder Input Log, our system includes built-in timeframes to provide feedback acknowledgement within 14 days and resolution within 60 days. The community engagement manager for the TIRE site will be responsible for tracking and responding to stakeholder feedback or delegating responses to other team members where appropriate (e.g., for in-person engagement). All field staff at the TIRE facility will also be made aware of the feedback tracking mechanism so that critical feedback conveyed during a stakeholder conversation can be recorded. We will share an updated Stakeholder Input Log with Isometric every quarter.

6. Integration with Grievance Mechanism

Our grievance mechanism is integrated with the Stakeholder Input Log described above and is in compliance with the Isometric Standard. The process for resolving grievances is as follows:

- Within 14 days of grievance received, Vaulted will send an email in acknowledgement to the stakeholder including the email of a point of contact on External Affairs. Also within 14 days, Vaulted will email Isometric with details about the grievance.
- Depending on the severity of the grievance, Vaulted will meet with the stakeholder(s) to collect more information and include summaries of these meetings in its Stakeholder Input Tracker.
- Depending on the type of grievance, Vaulted will organize a cross-functional team and contract additional legal or community outreach support as needed to develop a response and resolution. Vaulted will in good faith evaluate community concerns supported by verifiable data points and, where feasible and compliant with permits and safety requirements, may adjust operational practices to reduce material adverse impacts.
- Within 60 days, Vaulted will develop a resolution plan to be shared with Isometric and the stakeholder.

General guidelines that Vaulted will follow to apply the grievance mechanism:

- Stakeholders adjacent to the project or waste routes experiencing direct impacts will be given priority for grievance resolution
- All stakeholder nuisance complaints will receive a response including if they are escalated to a grievance
- Responses should be delivered in the format that the stakeholder prefers (email, phone call, etc.)

7. Reporting and Transparency

Vaulted will submit an annual Stakeholder Engagement Report to Isometric that provides a summary of engagement at all active sites, which will be available to stakeholders upon reasonable request. This report could include a highlight of local engagement activities, a summary of FAQ we received from the community, an overview of engagement for the following year, and our contact information.

8. Monitoring and Continuous Improvement

Vaulted will review the SEP annually for effectiveness and update it as needed.

FILES

Community Engagement History.png, Final ApprovedTIRE_CEQA_Comments.pdf, Exemption Note_ TIRE April 2026 Validation and Verification.docx, TIRE stakeholder Map - Isometric submission 5-4-26 1.xlsx

D.2

Projects must outline the mechanism for stakeholders to voice, process and resolve grievances.

SUBMISSION

Vaulted is committed to inclusively and transparently engaging communities and stakeholders throughout the project lifecycle to ensure that site operations and biomass sourcing activities are informed by local feedback. While community engagement will always be tailored to local conditions and needs, this document provides the values, commitments, procedures, and frameworks guiding our decision making in alignment with the Isometric Standard Section 3.5.

As a fast-growing start-up, Vaulted knows that there is always room for improvement, especially the evolving field of community engagement. Earlier this year, we conducted a literature review and interviewed 13 different stakeholders to identify best practices and common pitfalls related to stakeholder engagement, local permitting, community benefits and investment, and the procedures and resources needed to execute on these. By combining these learnings with our early engagement experiences, we identified the need to hire additional staff, stand up a stakeholder tracking and grievance resolution process, develop a community engagement playbook, and differentiate between permitting and engagement workstreams.

Summary

Vaulted uses a systematic and replicable stakeholder mapping process during early project development. This includes:

- Conducting preliminary research on local environmental, economic, demographic, and political conditions.

- Defining geographic scope based on regional ties and environmental vulnerabilities.
- Identifying stakeholders through desktop research, local media, consultants, landowner meetings, and personal connections.
- Categorizing stakeholders by type (e.g., community members, government bodies, NGOs, educational institutions).
- Special focus is placed on fenceline stakeholders and stakeholders directly impacted by the project.

Stakeholder Mapping

In the initial stages of project development and site selection, Vaulted spends time learning about the community and its surrounding region to improve Vaulted's understanding of local environmental, economic, demographic, and political characteristics—past and current. This preliminary research helps define the geographic scope for stakeholder mapping and identifies regional cultural or environmental factors. This is particularly important for regions reliant on fragile natural resources or in communities that have historically experienced high levels of environmental burdens. Vaulted uses the following methods to inform this preliminary step:

- Desktop research
- Local news and social media monitoring
- Contracting with a local consultant
- Meetings with potential landowners, or
- Outreach to existing connections in the area

Preliminary research enables Vaulted to develop a more in-depth understanding of the local community and broader regional connections, and in the process Vaulted naturally begins to identify stakeholders. This preliminary research was conducted during the permitting process for the TIRE facility.

Stakeholder maps are built and developed around organization type, such as community members (landowners, farmers, local residents, waste partners), governmental bodies (county and municipal government, permitting authorities, regional planning bodies), economic development organizations, educational partners (community colleges, local unions), and local non-profits and regional NGOs (environmental, community development, farmer's associations). Vaulted pays special attention to identifying stakeholders that could be directly impacted by our projects at a local level.

3. Engagement Methods and Frequency

Summary

Vaulted solicits feedback and assesses community needs through listening tours and local events. Engagement is tailored to each site's context, considering cultural norms, literacy, and community history. Vaulted will hold community office hours at TIRE to answer questions about the new project.

Stakeholder Outreach

Listening tours are fundamental to understanding community needs, and they will also help Vaulted understand which kinds of community engagement activities are appropriate. For projects in areas that are more densely populated, are greenfield sites, or which have a higher level of community scrutiny, selected engagement activities could include hosting office hours with subject matter experts or targeted surveys to reach a wider audience. Additionally, if the area has a high percentage of non-native English speakers, Vaulted could host a booth at a local cultural festival with translated materials and interpreters.

4. Input Collection and Tracking

Summary

Vaulted collects and tracks stakeholder input via:

- Site-specific email addresses, phone numbers, in-person meetings, public hearings, permit comments, open houses, and technical workshops.
- A formal Stakeholder Input Log (via Monday.com) that tracks feedback, concerns, actions taken, and resolution status.

Feedback Solicitation

Vaulted is committed to providing easily accessible and ongoing opportunities for community stakeholders, regulators, and waste suppliers to provide meaningful feedback.

Vaulted's Community Engagement Playbook (see attached) outlines timelines for engagement activities aligned with our work phases (from 'identify sites' through 'operations'). The 'project development' phase doesn't officially begin until construction starts following final permit approvals, but we begin collecting feedback and incorporating it into design changes throughout the permitting process. The permitting process requires engagement with strategic partners throughout and as well as assessing comments collected during public comment periods up until final designs receive all final permit approvals. When the project enters the 'operations' phase, Vaulted continues to collect feedback throughout the project's lifespan.

In order to solicit feedback from all potential community stakeholders, all Vaulted sites have a site-specific email address launched during the project announcement and administered by External Affairs team members. In addition, all facilities have a phone number for on-site inquiries.

Vaulted will create a site-specific email address and phone number for the TIRE facility.

As mentioned in sections above, additional engagement activities to build relationships and solicit ongoing feedback include: one-on-one meetings with stakeholders, attending public hearings, monitoring public comments during permitting, hosting open houses, hosting technical workshops, etc.

Alignment with Isometric Standard

For every project site, Vaulted holds a community open house in alignment with permitting timelines and Isometric's guidelines for accessibility in community consultation meetings:

- Stakeholders and rights-holders must be invited to consultation meetings with a minimum notice of 14 days.
- Stakeholders and rights-holders must be invited to consultation meetings via appropriate methods, which may include, but are not limited to, the post, email, or notices in newspapers and public places.
- Consultation meetings must be scheduled to maximize attendance, taking note of cultural or religious holidays and heritage.
- Meetings, documentation and correspondence must be in the local language(s), or have a translator where necessary to facilitate communication.
- Meetings, documentation and correspondence should be respectful of local knowledge; and accessible to a non-technical audience.

Recording Stakeholder Feedback

Vaulted recently stood up its stakeholder input log. Stakeholder maps will be built in the input log and then used to document ongoing interactions as well as incoming engagement.

The types of feedback recorded in the stakeholder input log will include concerns received via email interactions, outcomes of External Affair's stakeholder meetings, and questions received from our online contact form. As a small team, we often rely on staff integrated within the community on the ground. Every month, External Affairs plans to send an email to local staff and contractors requesting reports of any community concerns or other sentiments they have observed. The email will also remind them of the availability of the grievance mechanism.

The input log documents key details about the stakeholder, our relationship with them, preferred communication methods, as well as their input and sentiment toward the project. The input log is also

integrated with our grievance mechanism process in that it tracks our timelines for acknowledgement, resolution, and closure status. This will enable both tracking of feedback and escalation of grievances to be monitored and reported in a more robust and organized way.

5. Response and Follow-Up

Summary

Vaulted commits to:

- Acknowledging feedback within 14 days and resolving it within 60 days per the Isometric Standard.
- Reporting grievances to Isometric within 14 days and sharing quarterly updates with Isometric.

Feedback Tracking and Response Mechanism

As demonstrated in the Stakeholder Input Log, our system includes built-in timeframes to provide feedback acknowledgement within 14 days and resolution within 60 days. The community engagement manager for each project site will be responsible for tracking and responding to stakeholder feedback or delegating responses to other team members where appropriate (e.g., for in-person engagement). All Vaulted field staff will also be made aware of the feedback tracking mechanism so that critical feedback conveyed during a stakeholder conversation can be recorded. We will share an updated Stakeholder Input Log with Isometric every quarter.

6. Integration with Grievance Mechanism

Summary

Vaulted distinguishes between nuisance complaints and formal grievances:

- Grievances involve direct and verifiable impact or harm (e.g., road damage, illness, significant odor).
- Repeated nuisance complaints may escalate to grievances.

Grievance process includes acknowledgment of concerns, stakeholder meetings, cross-functional response planning, and resolution within 60 days.

Grievance Mechanism

Vaulted is committed to transparency and providing stakeholders with the opportunity to voice grievances and concerns. Stakeholder feedback, including grievances and complaints, can be received from a variety of sources, such as via e-mail, phone calls, in-person events, comments on permit applications, or an online contact form. Depending on the characteristics of the community, Vaulted uses different methods to improve accessibility and promote opportunities to provide feedback. Some examples include putting our contact information on flyers and invitations for local meetings or in other languages where needed.

Vaulted differentiates between two types of negative feedback from directly impacted stakeholders: nuisance complaints and grievances. Nuisance complaints include things like one-time issues with smells or noise. A nuisance may be escalated to a grievance when there is evidence that our project is causing direct, ongoing, and verifiable harm or negative impact, such as road damage, illness, a debris spill, etc.

While Vaulted will respond to all stakeholder feedback, nuisance reports will only be escalated to "grievance" level after we receive a sufficient number of complaints about an issue to establish a pattern and suggest causality tied to Vaulted's operations. For example, if we receive a few complaints about a smell issue, these initial complaints could just be a biproduct of TIRE's location adjacent to a waste water treatment plant. However, if we receive repeated complaints about smell from several different stakeholders directly adjacent to a facility, this establishes a clear pattern of negative local impacts, and this issue would be elevated to a grievance. At that point, the grievance mechanism described below would kick in.

Our grievance mechanism is integrated with the Stakeholder Input Log described above and is in compliance with the Isometric Standard. The process for resolving grievances is as follows:

- Within 14 days of grievance received, Vaulted will send an email in acknowledgement to the stakeholder including the email of a point of contact on External Affairs. Also within 14 days, Vaulted will email Isometric with details about the grievance.
- Depending on the severity of the grievance, Vaulted will meet with the stakeholder(s) to collect more information and include summaries of these meetings in its Stakeholder Input Tracker.
- Depending on the type of grievance, Vaulted will organize a cross-functional team and contract additional legal or community outreach support as needed to develop a response and resolution. Vaulted will in good faith evaluate community concerns supported by verifiable data points and, where feasible and compliant with permits and safety requirements, may adjust operational practices to reduce material adverse impacts.

- Within 60 days, Vaulted will develop a resolution plan to be shared with Isometric and the stakeholder.

General guidelines that Vaulted will follow to apply the grievance mechanism:

- Stakeholders adjacent to the project or waste routes experiencing direct impacts will be given priority for grievance resolution
- All stakeholder nuisance complaints will receive a response including if they are escalated to a grievance
- Responses should be delivered in the format that the stakeholder prefers (email, phone call, etc.)

Vaulted will submit an annual Stakeholder Engagement Report to Isometric that provides a summary of engagement at all active sites, which will be available to stakeholders upon reasonable request.

FILES

Duplicate_of_TEMPLATE_Stakeholder_Register_1759525450.xlsx, L - Community Engagment Playbook.xlsx, Exemption Note_ TIRE April 2026 Validation and Verification.docx, TIRE stakeholder Map - Isometric submission 5-4-26 1.xlsx

Section E: Pathway-specific

How will your project meet pathway-specific process requirements.

Monitoring requirements

E.1

Projects must create a table that outlines all monitored parameters in their selected protocol and modules.

SUBMISSION

The following table summarizes monitoring requirements for the facility as stated in the **Class V UIC Permit** for the TIRE facility .

Parameter	Parameter Description	Measurement Method	Monitoring Phase	Monitoring Frequency	Required Evidence	Isometric Module Reference
Injection Wellhead Pressure / MASIP	Maximum allowable surface injection pressure (MASIP) to prevent fracturing of confining layer. Continuous recording devices required. Automatic shut-off when limit exceeded.	Continuous pressure sensor / DAS with automatic shut-off alarm	Operation	Continuous	EPA Quarterly Reports; DAS Output	§4.1.1 Permeable Reservoirs Module v1.0
Injection Rate & Injected Volume	Maximum injection rate monitored to prevent induced seismicity or return of injectate. Continuous recording of injection rate and cumulative volume required.	Continuous flow meter / mass meter; DAS	Operation	Continuous	EPA Quarterly Reports; DAS Output	§4.1.1 Permeable Reservoirs Module v1.0
Injectate Temperature	Temperature of injection slurry. Required to characterise injectate and for input into reservoir/fracture simulation models.	Temperature probe/sensor (calibrated)	Pre-Injection & Operation	Continuous	EPA Quarterly Reports; DAS Output	§4.1.1 Permeable Reservoirs Module v1.0
Injectate Density	Density of injection slurry relevant to flow behaviour and migration modelling.	ASTM D 4052-22 or ASTM	Pre-Injection &	Daily	EPA Quarterly	§4.1.1; §4.2 Perme

Parameter	Parameter Description	Measurement Method	Monitoring Phase	Monitoring Frequency	Required Evidence	Isometric Module Reference
		D7042-21a (surface density); density at reservoir conditions calculated from T, P, salinity	Operation		Reports; Lab Evidence	able Reservoirs Module v1.0
Injectate Viscosity	Viscosity of injection slurry relevant to flow behaviour and migration modelling.	Industry standard viscosity measurement	Operation	Daily	EPA Quarterly Reports	§4.1.1 Permeable Reservoirs Module v1.0
Average Solids Concentration	Solids content of biomass slurry.	Industry standard gravimetric or optical solids analysis	Operation	Daily	EPA Quarterly Reports	§4.1.1 Permeable Reservoirs Module v1.0
Total Carbon © Content of Injectate	% weight of C in the biomass. Directly drives CO_2e removal quantification. Measured per Protocol §7.3.3.	TOC analyser; GC; or equivalent	Pre-Operation & Operation	Per batch	Lab Evidence (TOC); Loads (Integrated Evidence)	§4.1.1; Protocol §7.3.3 Biomass Geological Storage v1.1
Gas Detection — Wellhead / Facility	Gas detectors with alarms and automatic injection shut-off for CO_2, hydrocarbons or other gases with GWP>1. If triggered, must investigate immediately and report to VVB and regulatory body.	Gas monitors (≥0.01 vol% resolution); automatic shut-off system	Operation	Continuous	DAS Output; Alarm Logs	§4.1.1 Permeable Reservoirs Module v1.0
Internal Mecha	Testing of internal well mechanical integrity upon well completion, when well	Standard annular	Pre-Inject	Upon completion	MIT Report	§4.1.2 Perme

Parameter	Parameter Description	Measurement Method	Monitoring Phase	Monitoring Frequency	Required Evidence	Isometric Module Reference
nical Integrity Test	construction is modified, and at permit cadence. Includes annular pressure tests and corrosion assessment.	pressure test; corrosion inspection per API / ASTM	ion, Operation & Post-Injection	; when construction modified; then per UIC permit cadence performed annually	; UIC Permit; Testing Data	able Reservoirs Module v1.0
Annulus Pressure Monitoring	Continuous annulus pressure monitoring between injection tubing and long-string casing during injection. Pressure outside normal range triggers immediate investigation and reporting.	Continuous pressure sensor / DAS	Operation	Continuous	Continuous Monitoring Data; EPA Quarterly Reports	§4.1.2 Permeable Reservoirs Module v1.0
Pressure Fall-Off Test | External Mechanical Integrity Test	Quarterly pressure fall-off test to understand formation response to injection, evaluate induced fracture stability, and monitor injectate plume confinement (indirect method). Closure and fracturing pressures guide injection operations. Must be reported to Isometric, VVB and regulating authority if deviations from simulation occur.	Downhole pressure gauge; step-rate test (SRT) and pressure fall-off analysis	Operation	Quarterly	EPA Quarterly Reports; SRT/PFO Data	§4.1.2; §4.1.3.2 Permeable Reservoirs Module v1.0
Reservoir Pressure — Continuous	Continuous monitoring of reservoir pressure to demonstrate vertical containment within the storage complex. Distinct from wellhead/MASIP monitoring — this is downhole/formation pressure. Required for comparison to modelled predictions; major deviations trigger operational modification.	Downhole pressure sensor; DAS; or fibre-optic system	Operation & Post-Injection	Continuous	DAS Output; EPA Quarterly Reports	§4.1.3.2 Permeable Reservoirs Module v1.0

Parameter	Parameter Description	Measurement Method	Monitoring Phase	Monitoring Frequency	Required Evidence	Isometric Module Reference
Reservoir Temperature — Continuous	Continuous monitoring of reservoir temperature to demonstrate containment and detect unexpected changes. Temperature could be measured via fibre-optic DTS. Distinct from injectate slurry temperature.	Fibre-optic DTS; downhole temperature sensor/logger	Operation & Post-Injection	Continuous	DAS Output; EPA Quarterly Reports	§4.1.3.2 Permeable Reservoirs Module v1.0
Injectate Plume Confinement Monitoring	Periodic (e.g., monthly) indirect monitoring of injectate plume confinement and far-field containment. Results compared to reservoir model predictions to confirm the plume stays within the storage complex. Quarterly advanced pressure transient analysis also acceptable.	Pressure fall-off test (indirect); advanced pressure transient analysis; reservoir simulation comparison	Operation & Post-Injection	Monthly (indirect); Quarterly (pressure transient analysis)	EPA Quarterly Reports; Modelling Reports	§4.1.3.2 Permeable Reservoirs Module v1.0
Pre-existing Fracture & Fracture Development Monitoring	Step-rate tests (SRT) prior to injection to determine fracture breakdown and fracturing pressures. Regular PFO tests during operation to understand fracture stability. Fracture propagation must not intersect active faults. Deviations from simulation reported to Isometric, VVB and regulating authority.	Step-rate test (SRT); pressure fall-off (PFO); microseismic monitoring; well logs	Pre-Injection & Operation	Pre-injection (SRT); Regularly during operation (PFO)	SRT Data; PFO Reports; Monitoring Logs	§4.1.3.2 Permeable Reservoirs Module v1.0
Microseismic Monitoring	Required seismic monitoring program within AOR and storage complex for ≥M2.7 events to detect induced microseismicity. May use site-specific program or national program (e.g., USGS). Triggered re-evaluation if M2.7 or greater earthquake occurs within AOR.	Seismic network; microseismic sensors; or national USGS programme	Operation	Quarterly	Microseismic Reports; EPA Quarterly Reports	§4.1.3.2 Permeable Reservoirs Module v1.0
Surface Displacement / Subsidence	Surface displacement/subsidence monitoring to identify pressure changes and geomechanical impacts from injection. Compared to reservoir models to detect induced fracturing or reservoir volume change.	InSAR / SAR (satellite interferometry); tiltmeters; GPS instruments	Pre-Injection, Operation & Post-	Baseline (pre-injection); periodic during operation; per permit	Survey Reports; GPS / InSAR Data	§4.1.3.1 Permeable Reservoirs Module v1.0

Parameter	Parameter Description	Measurement Method	Monitoring Phase	Monitoring Frequency	Required Evidence	Isometric Module Reference
			Injection	post-injection		
Reservoir / Fracture Simulation Model	Mandatory reservoir modelling to estimate and locate injectate, gas generation and migration. Must include pressure and fracture simulations using industry-accepted software. Fracture simulation must provide estimates of fracture width, length and height. Compared to real monitoring data (T, P, porosity, permeability, injection history, mapped faults). Uncertainty analysis required.	Industry-accepted reservoir simulation software; fracture simulation; comparison to field data	Pre-Injection & Operation	Quarterly	Modelling Reports submitted to Isometric and VVB	§4.1.3.2 Permeable Reservoirs Module v1.0
Zone of Endangering Influence Evaluation	The ZEI study is performed to evaluate the areal extent of pressure in the injection zone and confirms that the current injection activity has no potential of causing vertical migration of injection fluid into the underground safe drinking water (USDW).	Updated reservoir model; site characterisation review	Operation	Quarterly	AOR Re-evaluation Report	§4.1.3.3; §4.1.3.2 Permeable Reservoirs Module v1.0
Biodegradation Process Evaluation — Gas & Formation Fluids	Samples of standing gas and formation fluids to quantify gas-water ratio, salinity, H_2, CO_2, H_2S, O_2, N_2, CH_4, ethane, and geochemical/biological properties. Required upon confirmation of gas presence from displaced brine or monitoring wells.	Wellhead/brine gas sampling; GC-MS; geochemical lab analysis	Operation	Quarterly	Lab Reports; Monitoring Logs	§4.1.3.2 Permeable Reservoirs Module v1.0; PDD

FILES

Monitoring Requirements.xlsx

Applicability

E.2

Projects must provide the address of the storage site to prove that it is in the United States or a region with similar geologic storage permitting requirements.

SUBMISSION

All biomass is stored and emplaced at the same Terminal Island site:

[redacted]

E.3

Projects must attest that they are not using biomass for enhanced hydrocarbon recovery.

SUBMISSION

Vaulted does not use biomass for enhanced hydrocarbon recovery. Wells at the TIRE facility do not contain oil or any other hydrocarbons.

E.4

Projects must provide evidence that the storage site has all required permits, including a valid UIC well permit, any environmental permits, and drilling, access, or land use permits.

SUBMISSION

The TIRE facility is regulated by the EPA under a UIC Class V permit, which was renewed in 2022. The Class V UIC permit is an 'umbrella' permit regulating all drilling and injection activities, so no additional land use or drilling permits are required. The site has also undergone review required by the California Environmental Quality Act (CEQA) in preparation for issuance of the EPA Class V permit, and the review found no impact or very minimal impact for all project activities. Therefore, no additional review or permitting was required under California law.

Vaulted submits quarterly monitoring reports to the EPA in line with Class V UIC permit requirements. Monitored parameters submitted in these quarterly reports are detailed in E.1. Vaulted's UIC Class V permit is publicly accessible here: **https://www.epa.gov/uic/R9UIC-CA5-FY18-1R**

FILES

Final ApprovedTIRE_CEQA_Comments.pdf

Environmental and social safeguarding

E.5

Projects must declare whether the biomass is non-hazardous or hazardous, based on U.S. EPA RCRA requirements and testing completed by an accredited laboratory.

SUBMISSION

The feedstocks processed at the TIRE facility are not regulated as hazardous wastes by the US EPA or the California Department of Toxic substances control. Biosolids from sewage treatment are not regulated as hazardous wastes under RCRA, and are instead **regulated under the Clean Water Act of 1972**.

In the Code of Federal Regulations Title 40, Part 503, the EPA establishes pollutant limitations, standards for pathogen reduction, management practices, and other provisions for biosolids, and the state of California **has adopted these minimum standards**. The majority of biosolids accepted at the TIRE facility are 'Class A' biosolids, which have been treated to remove the majority of pathogens present in the material. In emergency events such as extreme weather, TIRE may accept 'Class B' biosolids, which are treated to remove pathogens, but the treatment is not as extensive as for Class A biosolids.

While biosolids are not regulated as a hazardous substance under RCRA and biosolids are **regularly land applied, composted, or combusted**, the EPA has also found that biosolids can contain hazardous chemicals that can enter the environment when land applied.

NOTE to Isometric: Lab reports are forthcoming.

FILES

Lab Reports - HYperion Biosolids analysis.pdf

E.6

Projects must provide a risk assessment and mitigation plan for worker safety.

SUBMISSION

To ensure the health and safety of all personnel, Vaulted adheres to standard operating procedures for preventing and reporting incidents, conducting safety drills, and responding to heat stress and other job site hazards. See attached documentation.

FILES

T.I.R.E. Spill Prevention & Responce Plan.pdf, T.I.W.R.P. Emergency Responce Procedures.pdf, T.I.R.E. Daily Reporting Procedures & Guidelines.pdf, GET JSA Sheet QA-OP-013-2026-V2.MS updated.docx, GET Heat Stress Heat Index Management SOP_ JSA QA-OP-014-026-V2.docx

Biomass spills

E.7

Projects must provide a plan to monitor, document, and quantify biomass spills.

SUBMISSION

Any biomass spill occurring during transport between Hyperion and TIRE, or at either facility, will be managed in accordance with established best practices for safe biomass handling.

The volume of any lost biomass will be estimated based on the approximate proportion of the affected delivery load, cross-referenced against the relevant **delivery weigh ticket**. This is consistent with the protocol's mass measurement methodology, which uses the difference between truck arrival and departure weights on a calibrated scale to determine delivered mass (Section 7.3.3.1).

Where a spill results in a loss of biomass, the lost quantity will be **deducted from the delivered biomass quantity** for the relevant batch in the **GHG Statement**, as required by Section 7.3.3.3. Furthermore, if there is evidence that a biomass spill results in GHG emissions, those emissions will be captured under **CO_2eEmissions, n** and subtracted from the gross CO_2e stored in accordance with the net removal calculation

FILES

Spill Prevention & Responce Plan.pdf, Spill Response Plan.pdf

ISOMETRIC

Vaulted Deep

Feedstock: LACity_Biosolids

TIRE Project

Vaulted deploys slurry injection technology to geologically sequester organic wastes for the purpose of permanent carbon removal. The TIRE facility is a biosolids injection well co-located with the City of LA's Terminal Island wastewater treatment plant. Over the course of its operations over ten years, the site has safely and effectively injected more than 450,000 tons of biosolids. Vaulted has not yet generated any carbon removal units (CRUs) from TIRE – activities are currently supported through tipping fees from the City of Los Angeles. Vaulted has waived fees for waste volumes over the 5-year baseline to enable the city to send additional volumes to the site. Vault only credits those volumes over the baseline.

Feedstock requirements

A.1

Projects must provide a description of the feedstocks used.

SUBMISSION

The feedstock is a biosolid derived from the City of Los Angeles' waste water treatment process. The feedstock is sourced from the Hyperion Waste Water Treatment facility. The facility is owned and operated by the City of Los Angeles. [redacted], which is owned by the City of LA, or they are distributed to composting facilities that charge a tipping fee. [redacted]

A.2

Projects must list all organisations involved in providing the feedstock, with their company registration number, organisation name, address, contact person, contact email address, contact phone number and a description of the organisations role in the project.

SUBMISSION

Organization and registration number	**Vaulted Deep** [redacted]
Address	11000 Richmond Avenue, Suite 191, Houston, TX, USA
Contact person	Haley Leslie-Bole
Contact email address	haley@vaulteddeep.com
Contact phone number	(925) 639-0453
Description of role	Project proponent

Organization and registration number	**City of Los Angeles**
Address	200 N Spring St., Los Angeles, CA 90012
Contact person	[redacted] Please contact Vaulted first if communication with the City of Los Angeles is required.
Contact email address	[redacted] Please contact Vaulted first if communication with the City of Los Angeles is required.
Contact phone number	Call 311 or 213-473-3231

Organization and registration number	**City of Los Angeles**
Description of role	Waste supplier

FILES

CoLA Waste Partner Form_482026.docx, Final Staff Report for Carbon Credit Proposal with addenda.pdf

A.3

EC1: Project Proponent does not pay for the feedstock used.

SUBMISSION

Vaulted does not pay for biosolids feedstocks. Vaulted also receives no payment for this feedstock. Vaulted has waived tipping fees for volumes of waste exceeding 1,000 tons/week and receives no other payment for the biosolid feedstock. See the attached Carbon Credit Letter detailing the waste agreement and tipping fee waiver with the City of LA. This letter will be made publicly available as a 'Staff Letter' in 2026.

FILES

10.14.2025 Carbon Credit Letter Final.pdf

A.4

EC13: Project Proponent must assess the counterfactual fate of the feedstock.

SUBMISSION

The city of Los Angeles currently land applies the majority of biosolids from waste water treatment. Biosolids that will be used for this project are expected to be diverted [redacted] biosolids are land-applied and fodder for dairy cattle is produced.

Using a three-pool biosolids decay model, which assigns a literature-backed *k (decay) rate to a fast (labile) carbon pool, a slow (recalcitrant) carbon pool, and a mineral-associated organic carbon pool* (see attached document for methods and assumptions), Vaulted calculates remaining mass of C and CO2e at 15 and 50 years and uses this calculation to identify counterfactual emissions penalties per the Biomass Feedstock Accounting Module v1.2 Section 2.2. This method also takes a conservative approach to estimating the percentage of carbon that is emitted as methane in the labile and recalcitrant pools and also calculates counterfactual N2O emissions. This is then used to calculate CO2 equivalents.

Isometric directs project proponents to use the following approach to calculate 15 and 50 year counterfactual carbon storage. These values are generated using the model in the attached spreadsheet.

Using this analysis, Vaulted discounts 1% of embodied feedstock carbon before crediting. A small amount of the embodied carbon in biosolids is likely mineralized in soil and thus may remain in soil at 100 years.

FILES

NEW Isometric TIRE Biosolids Decay Model.xlsx

A.5

EC15: Feedstock is not used for energy production.

SUBMISSION

The City of Los Angeles does not incinerate biosolids or use biosolids to generate renewable natural gas via anaerobic digestion. Therefore, the feedstock is not used for energy production. See attached.

FILES

TIRE Quota Data - 2025.xlsx

A.6

Projects must describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock.

SUBMISSION

The City of Los Angeles land applies approximately 71% of its biosolids, composts 12%, and injects 17% at the TIRE facility (**Biosolid Uses, City of LA**). Land application takes place at [redacted] which is owned by Los Angeles Sanitation and has been a land application site since 1994. The site grows grain crops such as wheat and millet which are sold to local dairies as fodder.

The attached spreadsheet shows 2021-2025 volumes of biosolids that have been land applied, delivered to compost facilities with a tipping fee, and injected at the TIRE facility.

Vaulted's involvement will likely divert biosolids from land application and use as a soil amendment [redacted] City of Los Angeles has attested to

the fact that it will not replace diverted biosolids with synthetic fertilizer (see email screenshot). Accordingly, Vaulted does not account for replacement emissions.

FILES

TIRE Quota Data - 2025.xlsx, Fertilizer Replacement Email Screenshot.png

A.7

Projects must outline the method of replacement emissions accounting selected for this feedstock. Feedstocks must either be accounted for in the LCA or projects must explain how they are exempt under C1, C2, or C3.

SUBMISSION

[redacted] is a waste disposal site specifically operated to receive waste biosolids from the City of Los Angeles and crops are produced to provide supplemental income to the disposal facility.[redacted], a Los Angeles city official has attested to the fact that [redacted] will not purchase synthetic fertilizer to replace diverted biosolids (see attached).[redacted] Given that this site is primarily a waste disposal site, the City of Los Angeles has no reason to pay for fertilizer. The attached market leakage assessment also indicates that any change in crop production at [redacted] will have negligible market leakage impact.

FILES

cba2016-gilbert_jones_healthy_soils_from_biosolids_2016-11.pdf, Fertilizer Replacement Email Screenshot.png, CoLA Waste Partner Form_482026.docx, [External] Vaulted TIRE Biosolids Market Leakage.docx